

L250 Griffin EVK

OCI MSA DWDM Link

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July 21, 2026

Summary

We can accept [MRD request](#) that nominal power in laser delivery fiber is 21dBm for L250 EVK operating at OCI gen2 rate.

We caution that the dB's of margin is less than dB's of uncertainty. Numbers should be taken as +-2dB uncertainty at this stage.

This is the L250 EVK view, reusing forecasts from previous activities assuming serdes that has 2 pre-cursor taps, 2 post-cursor taps and 1 DFE tap

It is definitely not a forecast for L250 product -- those numbers are months away.

The $\geq$ symbol in the "Nominal power in laser delivery fiber, single λ (dBm)" column indicates that we can tolerate more powerful lasers also. We would use Tx VOA to manage excessive power, if necessary. Up to some practical limits which we have not yet specified. Therefore a laser suitable for OCI gen2 in this project would also be suitable for OCI gen1.

Finally, note that the MRM OMA numbers used correspond to what was used on Endeavor. It is possible that the OMA numbers are worse. At this point, there is no additional information or simulation available to put in definitive numbers.

OCI MSA wavelength bands

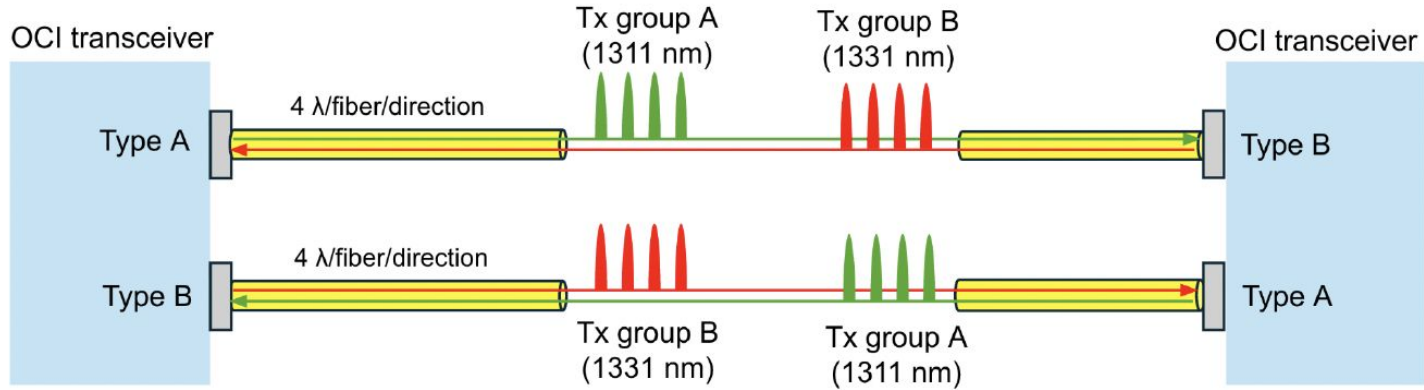


Figure 2-1. Illustration of DWDM wavelength channels and CWDM wavelength groups for a bidirectional link.

Parameter	Symbol	Min	Typical	Max	Unit	Test condition
Group A wavelength 0	WL0_TXA	1307.8	1308	1308.2	nm	PRBS13
Group A wavelength 1	WL1_TXA	1310.08	1310.28	1310.48	nm	PRBS13
Group A wavelength 2	WL2_TXA	1312.38	1312.58	1312.78	nm	PRBS13
Group A wavelength 3	WL3_TXA	1314.68	1314.88	1315.08	nm	PRBS13
Group B wavelength 0	WL0_TXB	1327.49	1327.69	1327.89	nm	PRBS13
Group B wavelength 1	WL1_TXB	1329.85	1330.05	1330.25	nm	PRBS13
Group B wavelength 2	WL2_TXB	1332.21	1332.41	1332.61	nm	PRBS13
Group B wavelength 3	WL3_TXB	1334.58	1334.78	1334.98	nm	PRBS13

OCI - MSA Gen1 Tx Specs

		Min	Typical	Max	Unit	
Total average launch power per group	Pavg_total			6	dBm	PRBS13
Average launch power, each channel ¹	Pavg	-8.5		0	dBm	PRBS13
Optical modulation amplitude (OMA), each channel	OMA	max(-5.5, -6.9+TDEC)		-1	dBm	PRBS13
Difference in launch power (OMA) between any two DWDM channels	dOMA			3	dB	PRBS13
Transmitter and dispersion eye closure (TDEC) ²	TDEC			3.4	dB	SSPR
Extinction ratio, each channel	ER	3.5	4.5		dB	PRBS13

OCI - MSA Gen1 Rx Specs

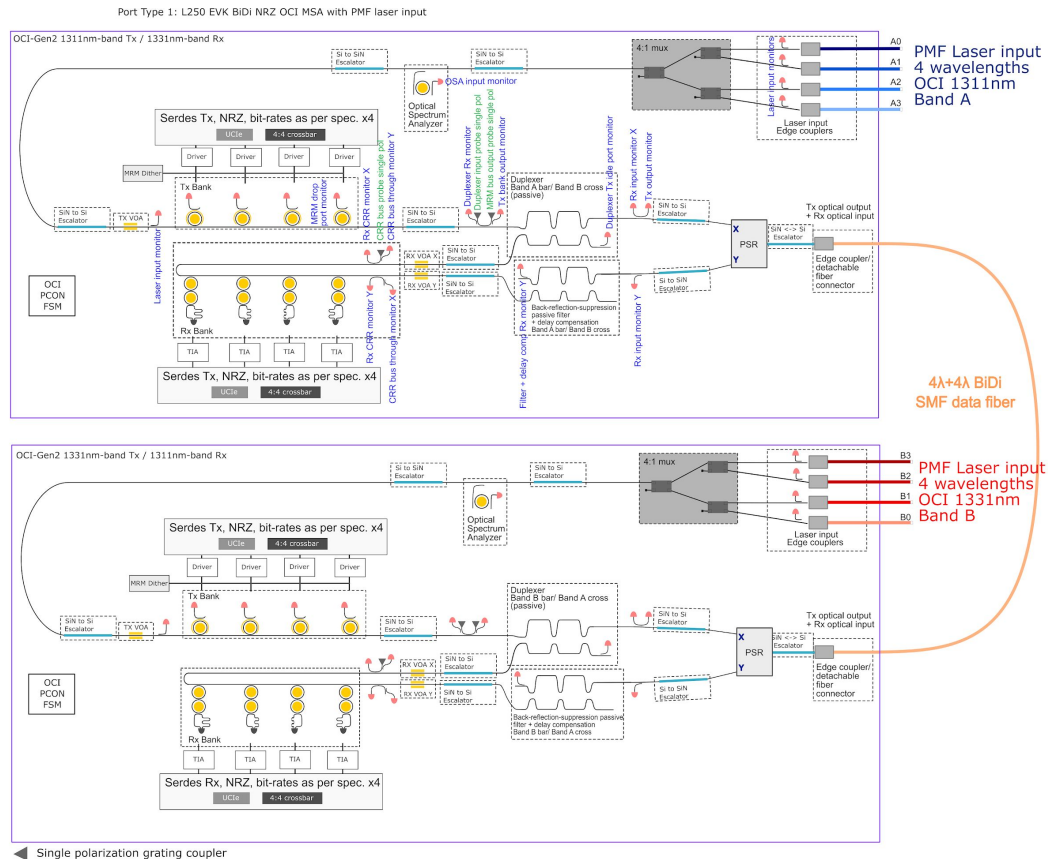
		Min	Typical	Max	Unit	
Damage threshold, each channel		4.5			dBm	
Average receive power, each channel ¹	Pavg	-11		0	dBm	
Difference in receive power (Rx OMA) between any two channels	dOMA			3	dB	
Receiver reflectance	Rx_Ref			-19	dB	
Optical modulation amplitude, each channel	OMA			-1	dBm	
Receiver sensitivity (OMA), each channel	RxSens			max(-8.2, -9.6 + TDEC)	dBm	PRBS31
Stressed receiver sensitivity (OMA), each channel ²	SRS			-6.2	dBm	PRBS31
BER floor ³	BER_FL			1E-6		

- Measured with conformance test signal at TP3 for the BER = 2.4E-4. The stressed conditions are shown in Table 2-4.
- Measured with a reference transmitter to produce TDEC greater than or equal to 2dB. The BER at receiver must stay within the specified limit over an OMA range of $(-8.2 + \text{TDEC})$ dBm to -1 dBm.

Photonics Block Diagram, with PMF laser

Key features

- OCI 4+4 bidirectional photonics
- Lightmatter photonic control IP
- PMF laser fiber
- Photonic layer signaling PIC to PIC
- WDM training, loss of signal, TRx connectivity ID



Laser power estimate

Based on inputs from Dominic⁷

Based on BER compliance at TP3 for BER of 1e-6 assuming TDEC of 2.5dB

Gen 1:

Min OMA at TP3 = $\max(-8.2 + \text{TDEC}) \rightarrow$ From OCI-MSA document, for BER 1e-6
= $\max(-8.2 + 2.5) = -5.7\text{dBm}$ (for Gen1)

Max. channel loss = 2.5dB.

Thus, Min OMA at TP2 = $-5.7 + 2.5 = -3.2\text{ dBm}$ (for BER of 1e-6)

Gen 2 (100G NRZ anticipated):

Min OMA (Gen 2) = Min OMA (Gen1) + 3dB (approximation) = -0.2dBm

Above is assuming worst case 3rd party RX. Based on prior sims by EOE team, it seems like we can achieve required BERs at much lower OMAs, using better Rx. Required OMA for these are estimated based on simulated BER vs OMA curves.

We then adjust laser power and a TX VOA (to control power into MRM, since MRM behaves badly under high power) until we meet above OMA @ TP2 for the -3σ link.

[Rollup link](#)

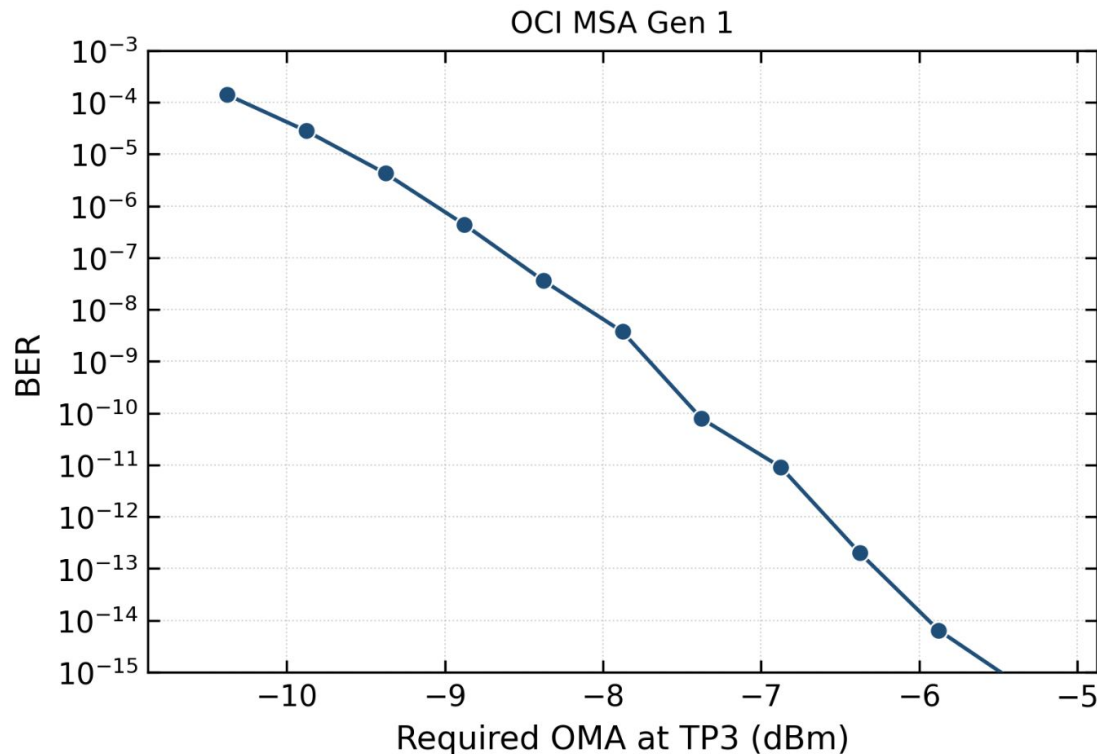
Courtesy: Alex

OCI MSA Gen 1 Required OMA

Rx Sensitivity

- Lightmatter has co-designed driver + MRM as well as PD + TIA for an OCI MSA Gen 1 deployment
- Design assumes Rx Serdes has:
 - 2 pre-cursor taps
 - 2 post-cursor taps
 - 1 DFE tap
- Includes worst case power penalties from spectral and coherent optical crosstalk

This graph is [Alex's simulation for OCI links for Nevada](#), based on Caribou DR OMNI Semi-Behavioral driver model from April 2026, and Caribou DR OMNI TIA from April 2026 [Note: these are not the latest Caribou DR models]



Laser power for Griffin L250 EVK OCI Gen1

Configuration	Laser split ratio in OE	Nominal power in laser delivery fiber, single λ (dBm)	Laser power range max-min(dB)	TX VOA clamp level (dBm)	Min TP2 OMA (dBm)	Min TP3 OMA (dBm)
LM Tx to worst case OCI compliant 3rd party Rx @ 1E-6	1:4	≥ 17	1.5 ^c	10	-3.2	-5.7 ^a
Worst case OCI compliant 3rd party TX to LM Rx	N/A	N/A	N/A	N/A	-4.4 ^a	-6.9 ^a Forecast BER ^b $\leq 5e-10$
LM Tx to LM Rx @ 1E-12	1:4	≥ 16	1.5 ^c	10	-4.1	-6.6 ^b

^a Based on OCI-MSA Gen1 standards document

^b Based on BER vs OMA curve simulated for Gen1

^c Min/Max is ± 0.75 dB from nominal and the choice of this range was somewhat arbitrary

Assumptions and Definitions:

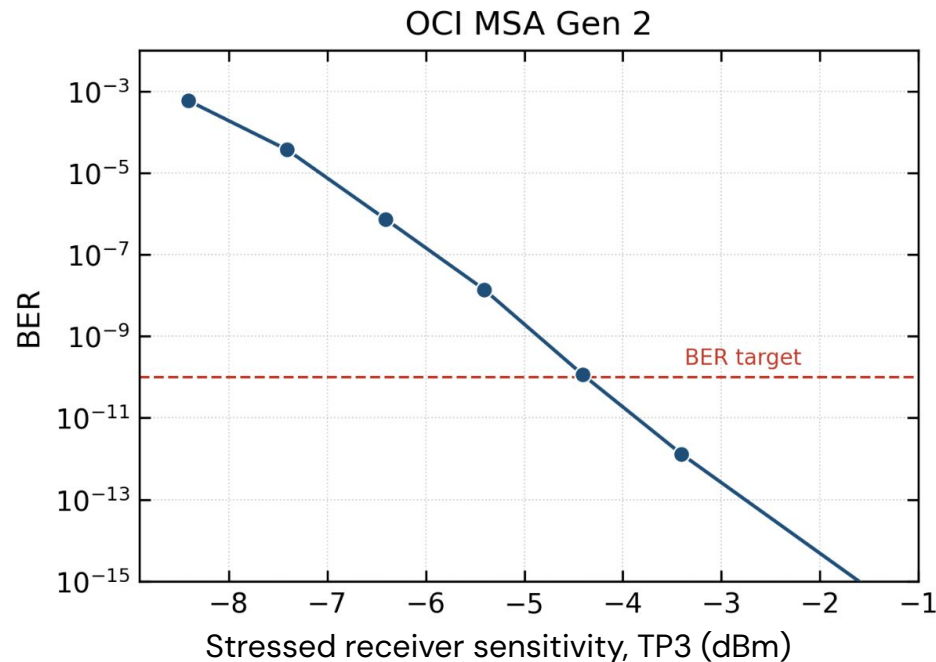
- Both LM and 3rd party Tx has TDEC of 2.5dB.
- relOMA of MRM is assumed to be 3.7dB
- 2.5 dB fiber channel loss OE to OE
- Split ratio = # laser ports : # TRx fibers
- Laser power = optical power in fiber from laser module

Courtesy: Alex

OCI MSA Gen 2 Required OMA

Rx Sensitivity

- Lightmatter has co-designed PD + TIA + Rx SERDES together for an OCI MSA Gen 2 deployment
- Design assumes Rx Serdes has:
 - 2 pre-cursor taps
 - 2 post-cursor taps
 - 1 DFE tap
- Includes worst case power penalties from spectral and coherent optical crosstalk



This graph is [Alex's simulation for OCI links for Nevada](#), based on Caribou DR OMNI Semi-Behavioral driver model from April 2026, and Caribou DR OMNI TIA from April 2026 [Note: these are not the latest Caribou DR models]

Laser power for Griffin L250 EVK OCI Gen2

Configuration	Laser split ratio in OE	Nominal power in laser delivery fiber, single λ (dBm)	Laser power range max-min(dB)	TX VOA clamp level (dBm)	Min TP2 OMA (dBm)	Min TP3 OMA (dBm)
LM Tx to worst case OCI compliant 3rd party Rx @ 1E-6	1:4	≥ 20.25	1.5 ^c	11	-0.2	-2.7 ^a
Worst case OCI compliant 3rd party TX to LM Rx	N/A	N/A	N/A	N/A	-1.4 ^a	-3.9 ^a Forecast BER ^b $\leq 5e-10$
LM Tx to LM Rx @ 1E-12	1:4	≥ 19.5	1.5 ^c	10	-0.9	-3.4 ^b

^a Based on adding 3dB (approximation) to OCI-MSA Gen1 standards document

^b Based on BER vs OMA curve simulated for Gen2

^c Min/Max is ± 0.75 dB from nominal and the choice of this range was somewhat arbitrary

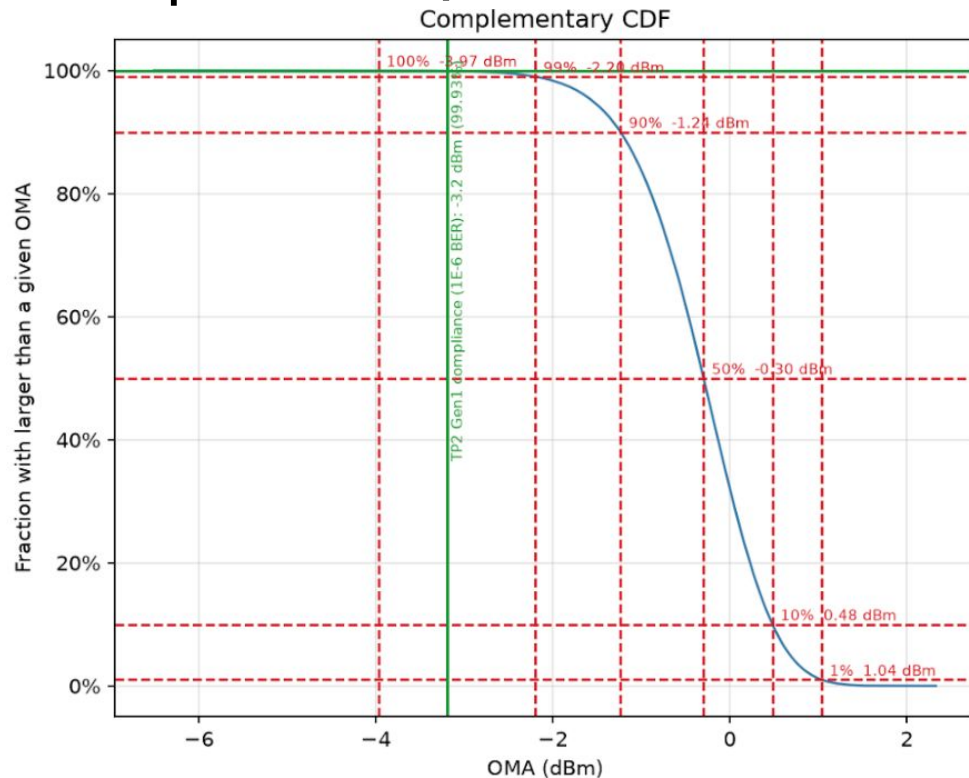
Assumptions and Definitions:

- Both LM and 3rd party Tx has TDEC of 2.5dB.
- relOMA of MRM is assumed to be 3.7dB
- 2.5 dB fiber channel loss OE to OE
- Split ratio = # laser ports : # TRx fibers
- Laser power = optical power in fiber from laser module

BACKUP

Yield curve for OMA at TP2, BER 1E-6, 50G NRZ (min OCI compliant Rx)

- Using PMF laser @ typ 17 dBm per lane
- 1:4 laser split
- 2.5 dB fiber channel
- Shown for worst OCI wavelength
- Slightly better with DWDM interleaved

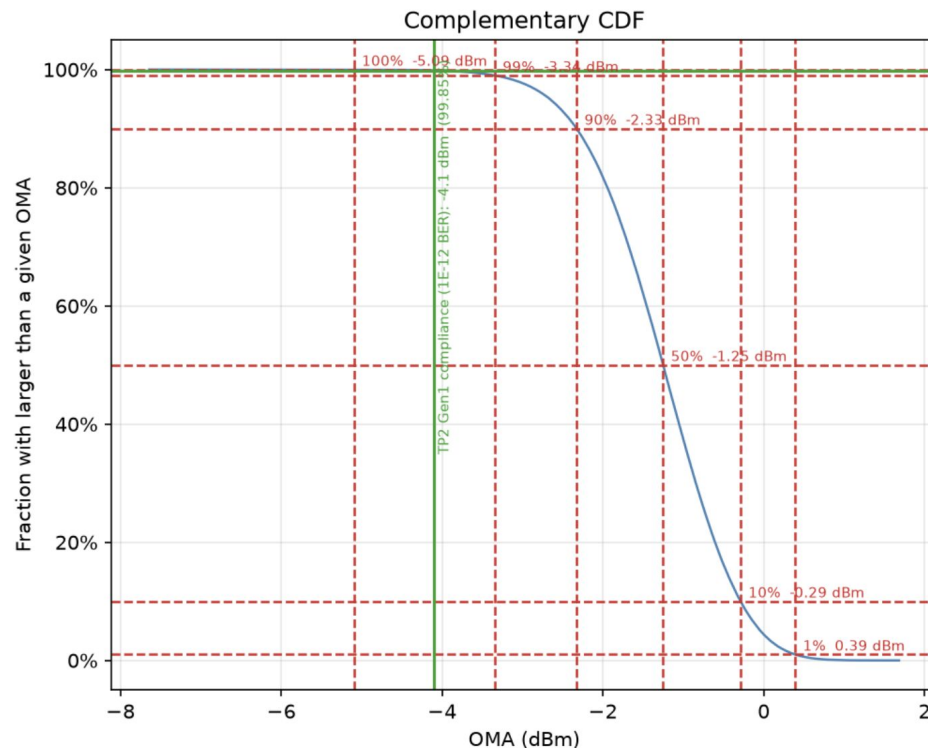


Notes

- Required OMA (dashed green line) corresponds to a BER of $1e^{-6}$
- Yield forecast is for one port. Port correlation / OE yield requires further analysis.

Yield curve for OMA at TP2, BER 1E-12, 50G NRZ (LM Rx)

- Using PMF laser @ 16 dBm per lane
- 1:4 laser split
- 2.5 dB fiber channel
- Shown for worst OCI wavelength
- Slightly better with DWDM interleaved

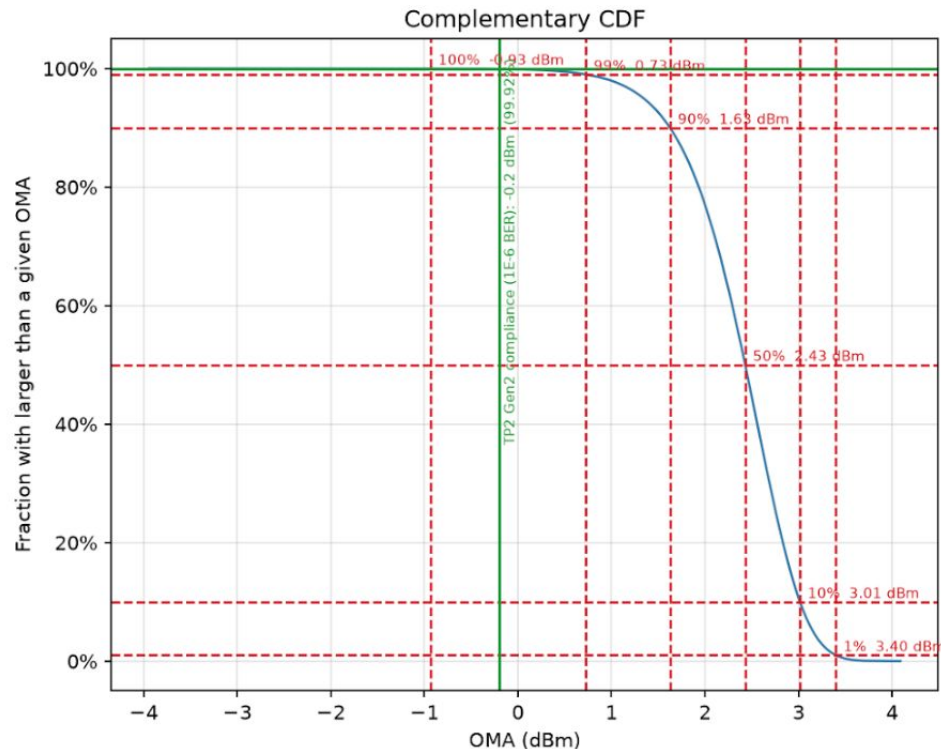


Notes

- Required OMA (dashed green line) corresponds to a BER of $1e^{-6}$
- Yield forecast is for one port. Port correlation / OE yield requires further analysis.

Yield curve for OMA at TP2, BER 1E-6, 100G NRZ (min OCI compliant Rx)

- Using PMF laser @ 16.5 dBm per lane
- 1:4 laser split
- 2.5 dB fiber channel
- Shown for worst OCI wavelength
- Slightly better with DWDM interleaved

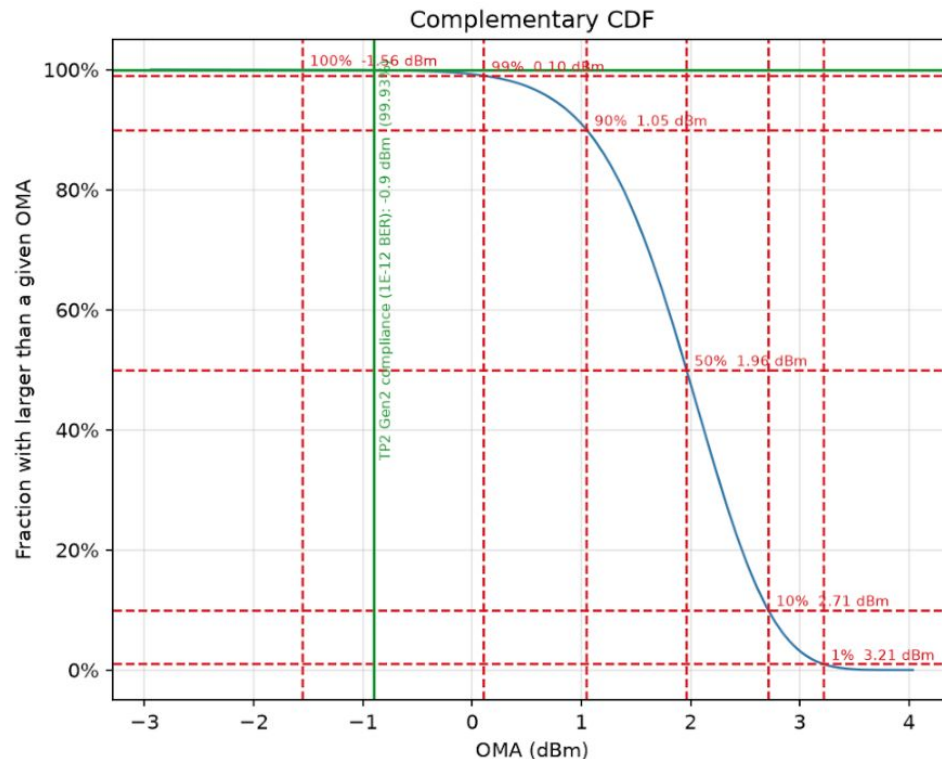


Notes

- Required OMA (dashed green line) corresponds to a BER of $1e^{-10}$
- Yield forecast is for one port. Port correlation / OE yield requires further analysis.

Yield curve for OMA at TP2, BER 1E-12, 100G NRZ (LM Rx)

- Using PMF laser @ 19.25 dBm per lane
- 1:4 laser split
- 2.5 dB fiber channel
- Shown for worst OCI wavelength
- Slightly better with DWDM interleaved



Notes

- Required OMA (dashed green line) corresponds to a BER of $1e^{-10}$
- Yield forecast is for one port. Port correlation / OE yield requires further analysis.

Why SM Laser delivery fibers not enabled

- NLA of pol locking circuit is too high. Refs: [SMF rollup for OCI](#), [PSR NLA data](#)
- Possible solutions
 - Develop nit only or low NLA PSR (photronics team has some initial designs)
 - Use Pol independent splitter to split light (1:8) first, then use rings or band mux to mux all lasers. Makes architecture and control very complex.